

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0001221	Version:	2.0, CURRENT
	Status:	Effective	Effective Date:	22-Jun-2011
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DG-57 / DG-517 Service Manual

PROBE X-Y FINE ADJUSTMENT IN PIPETTING POSITIONS

Procedure description

This procedure describes how to fine adjust probe X and Y coordinates on main pipetting positions. With this procedure absolute probe X-Y coordinates value are stored for :

- first card well in first position of Incubator rack;
- central position of dilution cup;
- centre tube of first samples carousel position;
- centre tube of first reagents carousel position;

Any other probe positioning is calculated relatively to those values.

Always perform the gripper adjust on incubator (REGD-0000807) before executing the probe X-Y fine adjustment (REGD-0001221).

Note: Document REGD-0001221_v2 replaces document REGD-0001221_v1.

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EQUIPMENT

1. A DG-57 instrument connected with a PC with the software installed.
2. Probe adjustment tool (P/N: 232170).

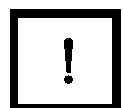
SET UP

1. Turn on the instrument.
2. Start the ST software (Technical Service software).
3. Enter the Technical Service password (d+g_diana).



WARNING! BIOHAZARD

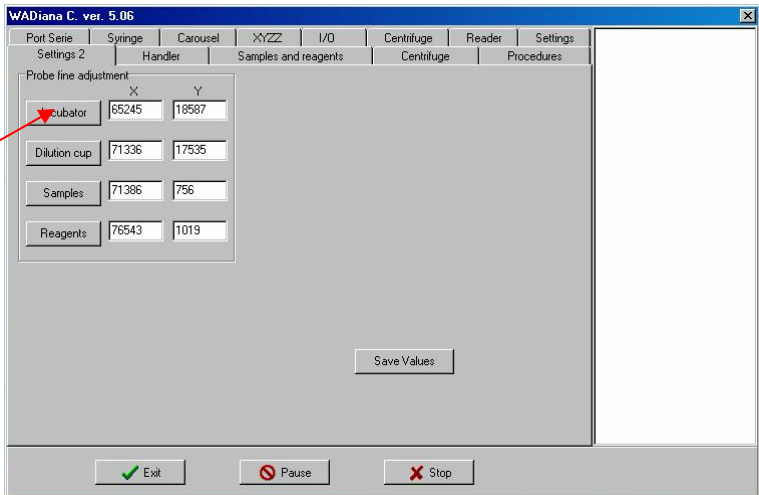
Handle with care, there may be rests of blood in the probe.
Clean the probe before executing the fine adjustment procedure.



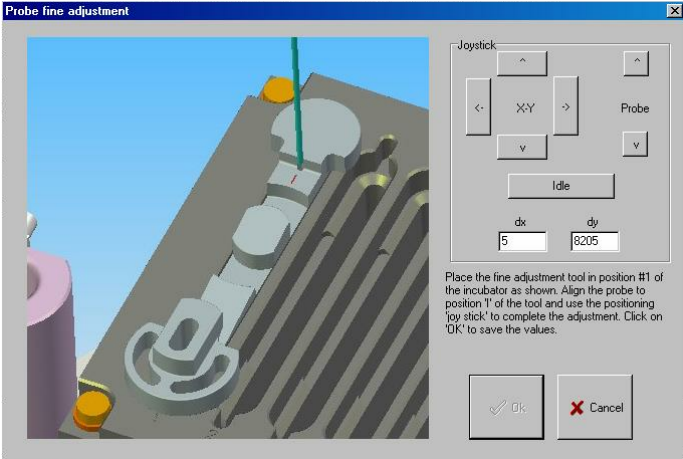
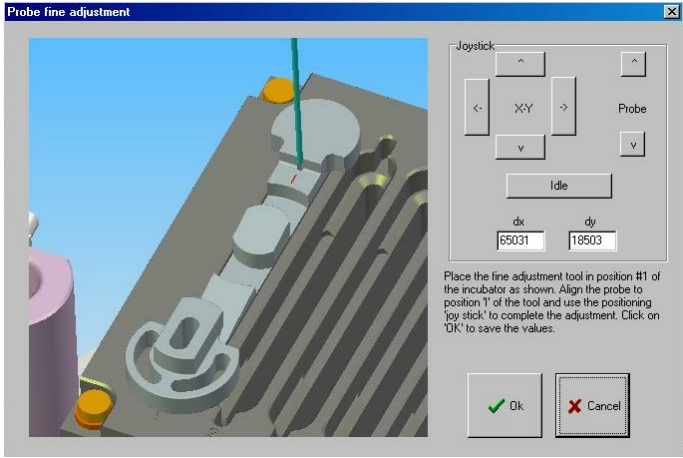
CAUTION!

Once a probe X-Y fine adjustment (*REGD-0001221*) has been executed for a specific equipment, executing again a gripper adjustment procedure (*REGD-0000807*) does not affect the values stored with the former procedure if the probe has not been changed. In case of probe changing both procedures must be repeated performing the gripper adjustment on incubator (*REGD-0000807*) before executing the probe X-Y fine adjustment (*REGD-0001221*).

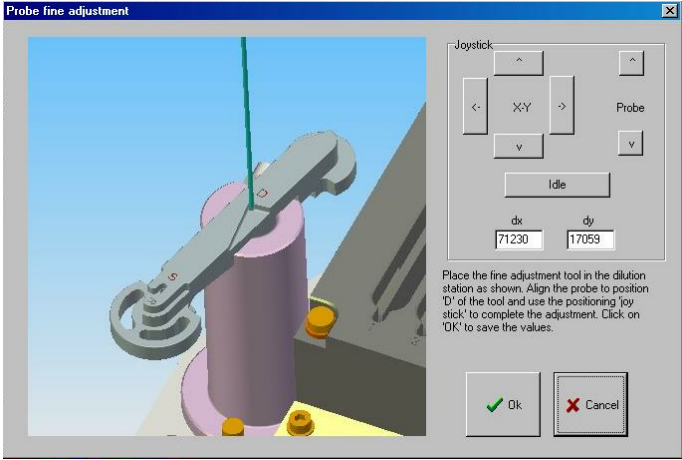
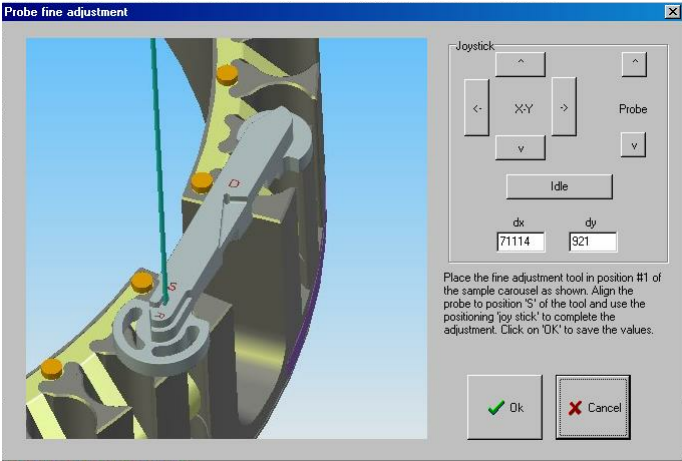
PROCEDURE

Process	Expected Results
1. Click on the “Settings 2” tab. Press on “Incubator” button in “Probe fine adjustment” screen.  Picture 1	New window appears (Picture 2)

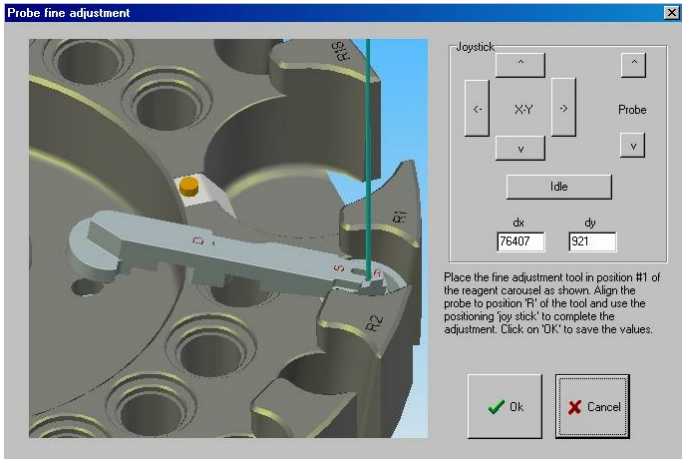
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Process	Expected Results
2. Place the tool as shown and move the arm and probe to fit the probe in the “I” hole of the tool. To speed-up procedure, the robot arm can be initially moved by hand after pressing the “Idle” button. To avoid X-Y positioning errors and block the motors use the joystick commands to introduce probe in the hole.  Picture 2	Ok button becomes active when probe X-Y coordinates are within a 3 mm square respect to the value calculated by the software after gripper adjustment procedure (Picture 3)
3. Once probe has been introduced to “I” hole, click on “Ok” button to return to “Settings 2” tab and store the new X-Y absolute values.  Picture 3	The probe automatically moves to home position and new values are shown in Incubator X-Y coordinates fields of “Settings 2” tab.
4. Press on “Dilution cup” button in “Probe fine adjustment” screen of “Settings 2” tab. 5. Place the tool as shown and move the arm and probe to fit the probe in the “D” hole of the tool. To speed-up procedure, the robot arm can be initially moved by hand after pressing the “Idle” button. To avoid X-Y positioning errors and block the motors use the joystick commands to introduce probe in the hole. Ok button becomes active when probe X-Y coordinates are within a 3 mm square respect to the value calculated by the software after gripper adjustment procedure. Once probe has been introduced to “D” hole, click on “Ok” button to return to “Settings 2” tab and store the new X-Y absolute values.	New window appears (Picture 4) The probe moves to home position and new values are shown in dilution cup X-Y coordinates fields of “Settings 2” tab.

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Process	Expected Results
 Picture 4 - Press on “Samples” button in “Probe fine adjustment” screen of “Settings 2” tab, samples carrousel automatically moves to place first tube lodgment in pipetting position. - Place the tool in first carrousel position as shown and move the arm and probe to fit the probe in the “S” hole of the tool. To speed-up procedure, the robot arm can be initially moved by hand after pressing the “Idle” button. To avoid X-Y positioning errors and block the motors, use the joystick commands to introduce probe in the hole. Ok button becomes active when probe X-Y coordinates are within a 3 mm square respect to the value calculated by the software after gripper adjustment procedure. Once probe has been introduced to “S” hole, click on “Ok” button to return to “Settings 2” tab and store the new X-Y absolute values.  Picture 5 - Press on “Reagents” button in “Probe fine adjustment” screen of “Settings 2” tab, reagents carrousel automatically moves to place first vial lodgment in pipetting position. - Place the tool in first carrousel position as shown and move the arm and probe to fit the probe in the “R” hole of the tool. To speed-up procedure, the robot arm can be initially moved by hand after pressing the “Idle” button.	New window appears (Picture 5). The probe automatically moves to home position and new values are shown in Samples X-Y coordinates fields of “Settings 2” tab. New window appears (Picture 6) The probe moves to home position and new values are

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Process	Expected Results
To avoid X-Y positioning errors and block the motors, use the joystick commands to introduce probe in the hole. Ok button becomes active when probe X-Y coordinates are within a 3 mm square respect to the value calculated by the software after gripper adjustment procedure. Once probe has been introduced to "R" hole, click on "Ok" button to return to "Settings 2" tab and store the new X-Y absolute values.  Picture 6 10. To export X-Y settings values to CalibValues.txt file (in Log folder) press "Save Values" button on "Settings 2" tab window. Be careful, this action overwrites existing CalibValues.txt file.	shown in Reagents X-Y coordinates fields of "Settings 2" tab. ---

DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	See document header.	Document REGD-0001221_v2 replaces document REGD-0001221_v1.	Document version changed.
2.0	See document header.	Pictures updated.	Pictures updated with Diana System software version 3.2 installation.
2.0	See document header.	Document title.	Commercial reference substituted by project reference.
2.0	See document header.	Document header.	Increase information to easy document search.